OPERATING SYSTEMS AND SYSTEMS PROGRAMMING - PROJECT POSTER

Team ID: 5 | Section: 9 | Members:

Members: B. Jhahnavi (2520030452),
P. Mouvya sri (2520030449) | **Guide:** Y harika devi

UNIVERSITY LABORATORY MANAGEMENT SERVER

Abstract

- Develops a centralized server-based system to manage university laboratory resources, handling user authentication, slot booking, and system activity logs using OS concepts like IPC and process management.

Problem Statement & Objectives

- **Problem:** Managing scattered laboratory resources leads to scheduling conflicts, inefficient resource tracking, and unauthorized access.

Objective 1: Multi-client server architecture.
Objective 2: Shared memory/message queues
Objective 3: File-based logging.

Existing vs Proposed System

EXISTING SYSTEM (Manual/Isolated)	PROPOSED SYSTEM (Automated Server)
• Manual resource tracking & scheduling	• Automated, centralized tracking
• Paper-based logbooks and tool booking	• Real-time, conflict-free booking
• Unauthorized resource access risk	• Secure, credential-based access
• Lacks centralized server & logs	• Automated server log generation

Feasibility Study

- **Technical:** C Programming, Linux/Ubuntu, Socket programming, IPC.

Economic: Open-source tools, no licensing costs.

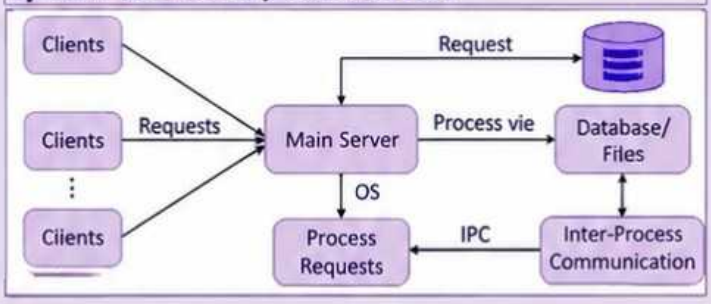
Operational: High reliability, secure access.

Schedule: Semester-timeline with milestones.

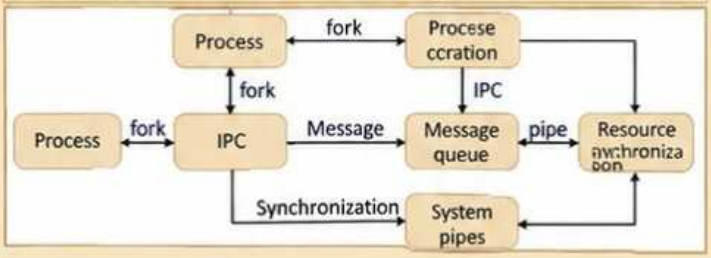
Tools & Technologies

- **Programming:** C (with multithreading &sockets)
- **Platform:** Linux / Ubuntu.
- **OS Concepts:** Proceses, threads, IPC, synchronization, file lo, king
- **System Calls:** socket(), fork(), exec(), bind(), listen()
- **Tools:** GCC, Make, Git & GitHub

System Workflow / Architecture



Process, IPC & Resource Design



Implementation Screenshots

Client Interface



Server Response



Results & Testing

- Testing includes client-server communication, multi-user concurrent access, and input validation, ensuring server stability.

Conclusion

Successfully implemented a multi-user bd server using how-leved OS primitives, demonstrating synchronization, process-isolation, and concurrent connections.

Future Scope

1. Adding a web/GUI interface.
2. Encryption for communication.

QR Code / Scan

References: [1] Linux man pages [2] Operating systems textbook [3] GCC / GNU documentation

GitHub: https://github.com/jhahnavibadugu27-hub/KLH-CSE-2026-Team5-UniversityLaboratoryManagementServer